

JECRC UNIVERSITY
Subject: Mathematics I (DMA024A)
Course: B. Tech-I Semester
Tutorial Sheet – 10

Unit-IV

M.M:64

CO4: To deal with functions of several variables that are essential in most branches of engineering.

Part A

(5×2=10)

A1. [CO4] State the condition under which a function $f(x,y)$ is continuous at a point (a,b) .

A2. [CO4] If $f(x,y) = e^{x+y}$. Find the value of $\frac{\partial^2 f}{\partial x \partial y}$.

A3. [CO4] If $z = x^3y + y^3x$. Find $\frac{\partial z}{\partial x}$ at $(1,2)$.

A4. [CO4] State whether the function $f(x,y) = |x| + |y|$ is continuous at the origin.

A5. [CO4] Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2+y^2}{x^2+y^2+1}$.

Part B

(3×7=21)

B1. [CO4] (a) Define the continuity of a function of two variables at a point.

(b) Test the continuity of $f(x,y) = \begin{cases} \frac{xy}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$ at the point $(0,0)$

B2. [CO4] For $z = e^{xy} + x^3y^2$, (a) Find $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$. (b) Verify $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$.

B3. [CO4] (a) Define the total derivative of a function of two variables.

(b) For $z = x^2 + xy + y^2$, find the total differential dz at the point $(x,y)=(2,-1)$ when $dx=0.01$ and $dy=-0.02$.

Part C

(3×11=33)

C1.[CO4] Let $f(x,y) = \begin{cases} \frac{xy}{x-3}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$

(a) Show whether f is continuous at $(0,0)$.

(b) If not continuous, show that along the line $y = mx$ the limit depends on m .

C2. [CO4] Let $u(x,y) = x^2y + e^{xy}$

(a) Find the first-order partial derivatives u_x and u_y .

(b) Find the second-order mixed partial derivative u_{xy} and verify Clairaut's theorem.

(c) Evaluate u_{xx} at the point $(1,0)$.

C3. [CO4] (a) Define the limit of a function of two variables. State the condition for the existence of a limit.

(b) Test the existence of the limit: $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2y}{x^4+y^2}$.

Answers

A2: e^{x+y} **A3:** $\frac{\partial z}{\partial x} = 3x^2y + y^3, \frac{\partial z}{\partial x(1,2)} = 14$ **A5:** 0

B3 (b): 0.03.

C2. (a) $u_x = 2xy + ye^{xy}, u_y = x^2 + xe^{xy}$ (b) Clairaut's theorem ($u_{xy} = u_{yx}$). (c) 0.